UTEMASTER OPERATIONS | QUALITY

POKA-YOKE VISION STATION

Laptop install and bench test

Step-by-step: install on a Windows laptop, build a 5-part test board, teach it, then run 21 checks to verify the full system works.

Time needed: about 2 hours. Date: 13/10/2026

What this proves

By the end you will have shown, on your own laptop and camera, that the station: finds every part on a real board, counts loose fasteners, catches a wrong-coloured part, ignores a hand in a pocket, copes with dim and uneven light, refuses to judge a covered camera, rejects the wrong board, handles a secondary QC sign-off, reads barcodes with the camera, and logs every check against an order.

Stage	What you do	Time
A. Install	Python, download the software, first run, 2-minute smoke test with no camera	25 min
B. Set up	Camera above an A3 board, lights, 5 parts, print the labels (last page)	25 min
C. Teach	Learn mode: empty board, 5 parts one at a time, 3 reference shots	15 min
D. Test	21 scripted bench tests, tick each one	45 min
E. Sign off	Order log check, results table, optional Ops Hub launch test	10 min

What you need

Item	Detail	Ready
Windows laptop	Windows 10 or 11, 8 GB RAM, about 1 GB free disk, internet for the install	[ ]
USB webcam (recommended)	1080p with a clip or tripod thread, e.g. Logitech C920 / C922 / Brio. The built-in laptop camera works for the smoke test, but can't look straight down at a board	[ ]
Camera mount	Tripod, desk-lamp arm, monitor arm or a shelf, holding the camera 50-60 cm above the board, lens pointing straight down	[ ]
Light	Room light plus one desk lamp you can move and switch off (for the lighting tests)	[ ]
Board	A3 (420 x 297 mm) light-coloured foam board or cardboard, black marker	[ ]
5 test parts	1 black bracket - 1 black rubber seal strip (about 30 cm) - 1 clear zip bag with some cable inside - 6 black bolts or screws - 1 bundle of white cable ties	[ ]
Swap parts for tests	A white bracket (the white variant, or white card cut to the bracket's shape) - a white strip the size of the seal (paper or white tape) - 6 silver/zinc screws - 1 extra black bolt - a second, different board or sheet of card	[ ]
Printed labels	Print the last page of this guide at 100% scale (no 'fit to page')	[ ]
GitHub sign-in	The software is in a private repository. Sign in to github.com as joshmcintyre-sudo	[ ]

A. Install (25 min)

Step 1 - Install Python

1. Open **python.org/downloads/windows** and download the latest **Python 3.12** "Windows installer (64-bit)".
2. Run it. On the first screen tick **Add python.exe to PATH**, then click **Install Now**. No admin rights are needed for the default per-user install.
3. Check: press **Win + R**, type **cmd**, Enter. In the black window type the line below. It must print Python 3.12.x.

```
py -3 --version
```

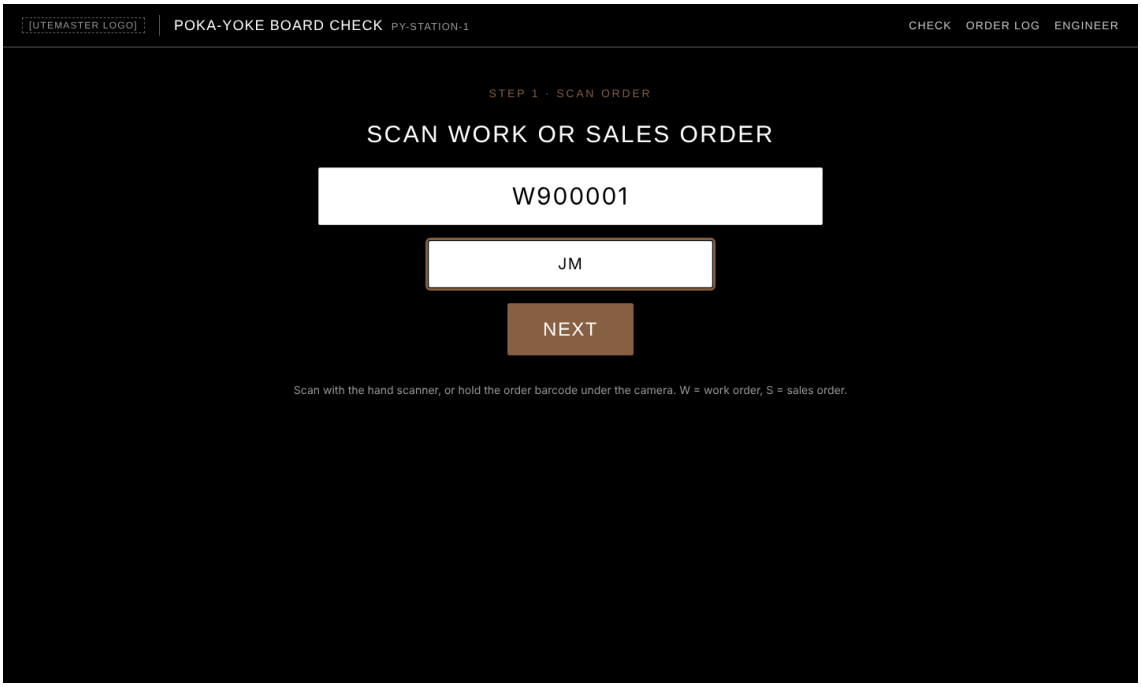
Step 2 - Download the software

1. Go to **github.com/joshmcintyre-sudo/Josh-Working** (signed in).
2. Click the branch drop-down (it says **main**) and choose **ccr-22c8396c-o54ml4**.
3. Click the green **Code** button, then **Download ZIP**.
4. Make a folder **C:\PokaYoke**. Right-click the ZIP, **Extract All**, and extract into C:\PokaYoke.
5. Open the extracted folder, then the **poka_yoke_vision** folder inside it. This is the station folder for the rest of this guide. It contains **run_station.bat**, **settings.json**, **README.md**.

Tip: copy the full path of the station folder from the Explorer address bar; you'll paste it in Step 4.

Step 3 - First run

1. Double-click **run_station.bat**. If Windows shows "Windows protected your PC", click **More info** then **Run anyway**.
2. The first run creates a private Python environment and downloads OpenCV, Flask and the barcode reader (2-5 minutes). Leave the black window open: it **is** the station. Closing it stops the station.
3. When the window shows **Operator screen: http://127.0.0.1:8080/**, the browser opens on the station. If the browser opened first and shows "can't reach this page", press **F5**.
4. You should see the black **SCAN WORK OR SALES ORDER** screen (picture below).



The first screen: order scan. Typing an order number works the same as scanning it.

Step 4 - Smoke test with no camera (2 min)

This proves the install works before any hardware is involved. It runs the built-in 5-part demo board from a picture with two problems in it.

1. Close the station's black window.
2. Open a command window in the station folder: in Explorer, click the address bar, type **cmd**, Enter.
3. Run these two lines:

```
.venv\Scripts\python -m pokayoke.demo
.venv\Scripts\python -m pokayoke.app --image demo_images\missing_parts.png
```

1. Open **http://127.0.0.1:8080/**. Type order **W900000**, initials, **Next**.
2. Pick **Hilux DC / Demo canopy fit-out kit [DEMO-HILUX-01]** from the list, then **VERIFY BOARD**.
3. **Expect:** ITEMS MISSING with **MISSING 1 x Door seal 450mm** and **Need 2 more M8 x 25 flange bolt (4/6)**. Press **CONFIRM & SAVE**.
4. Close the command window (this stops the demo). Smoke test passed: [ ]

Step 5 - Point the station at your camera

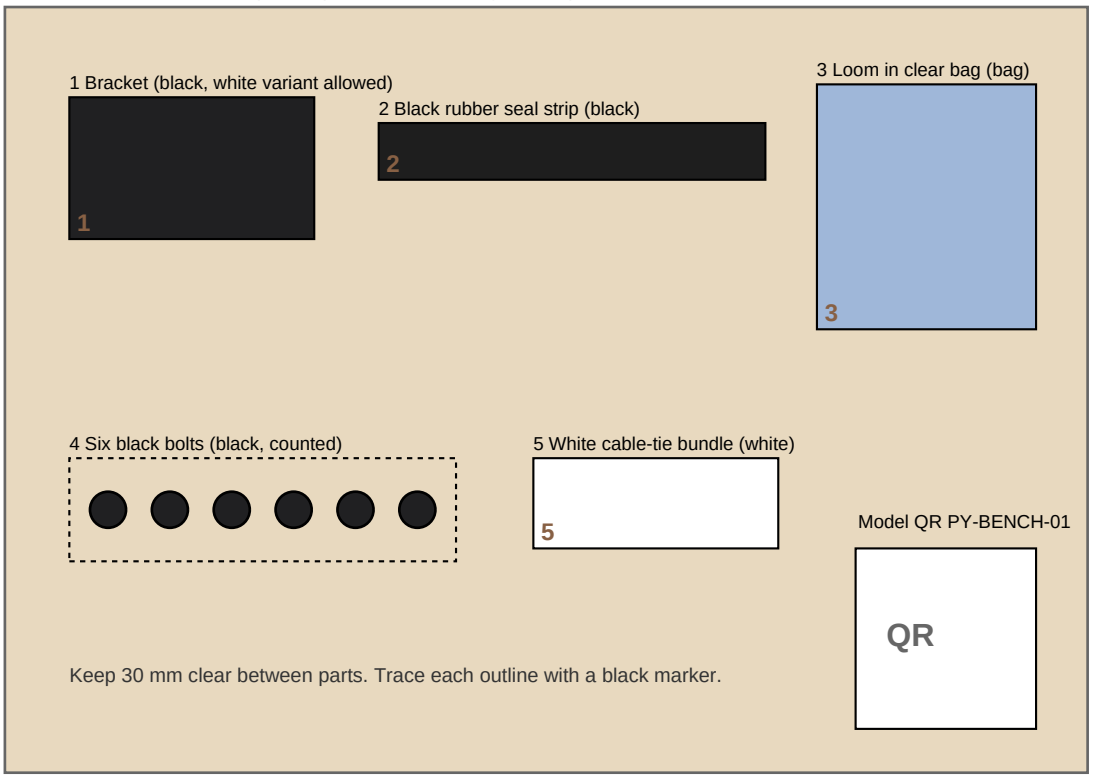
1. Plug in the USB webcam. Close Teams, Zoom or the Windows Camera app: only one program can use a camera.
2. In the station folder, right-click **settings.json**, **Open with**, **Notepad**.
3. Change these values, keep the commas and quotes exactly, then save:

Setting	Set to	Why
"camera_index"	1	0 = laptop's built-in camera, 1 = first USB camera. If the picture is from the wrong camera later, try 0 or 2
"station_name"	"BENCH-LAPTOP"	Shown on screen and in every result
"width" / "height"	1280 / 720 (leave)	1920 / 1080 only if parts look tiny on screen
"export" section	leave as is	Bench checks aren't launched from Ops Hub, so they stay on the laptop. The order log shows "not sent: standalone check". That is expected.

1. Double-click **run_station.bat** again. The station is ready.

B. Set up the camera and board (25 min)

A3 board 420 x 297 mm, top view (camera looks straight down)



Suggested layout. Any layout works, as long as parts don't touch and the whole board is in view.

#	Part	Material to choose	Qty	Why it's in the test
1	Bracket (black; white variant)	Black, also comes in White	1	Black 95% of the time, white occasionally: both pass, the white one is recorded
2	Black rubber seal strip	Black	1	Long thin part; swapped for a white strip to test WRONG PART
3	Cable in a clear zip bag	Bag	1	Shiny, changes shape every time it's placed
4	Black bolts or screws	Black (Steel if yours are zinc)	6	Counting: short, extra, fingertip
5	White cable-tie bundle	White	1	White on a light board: hardest contrast, glare test

Board and labels

1. Lay the 5 parts out as above with at least 30 mm between them. Trace round each one with the black marker (a shadow board). The marker lines also help the camera lock on to the board.
2. Write **PY-BENCH-01** in large letters in one corner.
3. Cut out the **model QR (PY-BENCH-01)** from the label page and stick it flat in the bottom-right corner.
4. Keep the order labels, the QC badge label and the other labels loose for the tests.

Camera and light

1. Mount the camera **50-60 cm above the board, pointing straight down**, centred over it.
2. In the browser open **http://127.0.0.1:8080/teach** and click **Live preview**.
3. Move the camera or the board until the **whole board plus about 2 cm of desk on every side** fills the picture. Focus should be sharp on the marker lines.

4. Turn on the room lights and put the desk lamp to one side at about 45 degrees, not pointing into the lens. Avoid direct sun on the board.
5. Fix the board's position with two strips of tape at its corners (locating stops).
6. Click **Stop preview**.

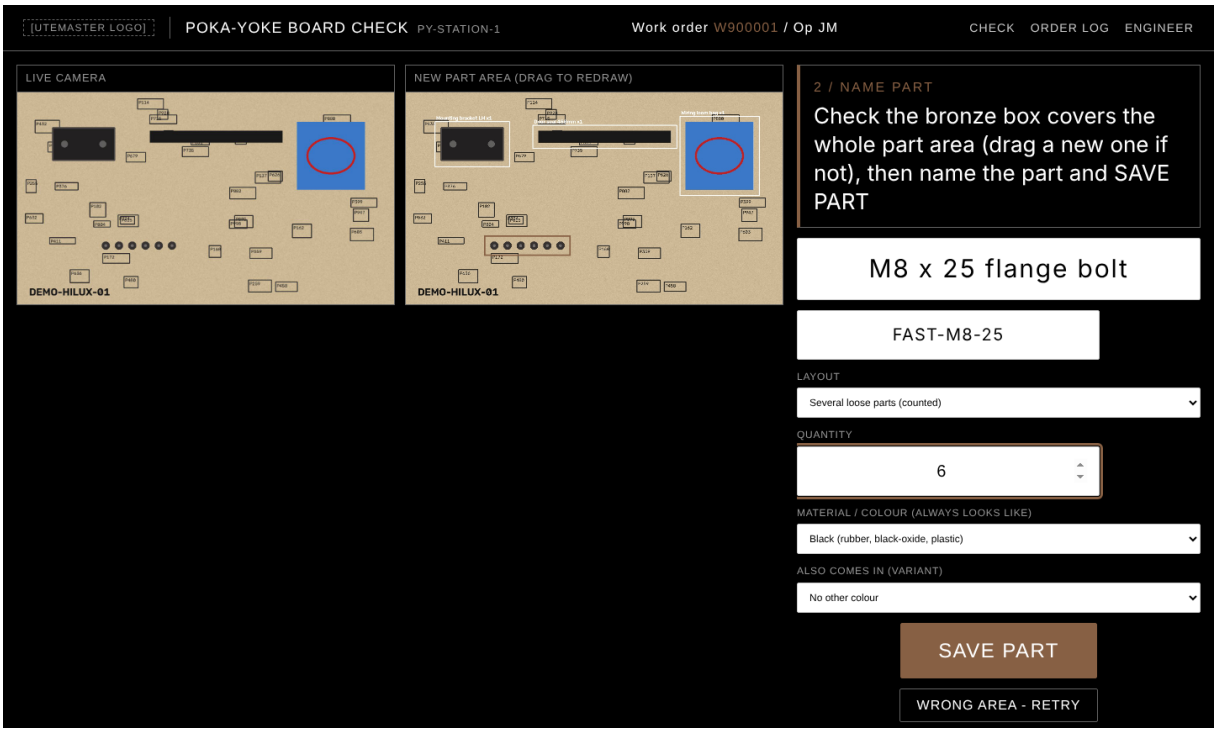
Rule of thumb: on screen, a bolt head should be at least about 20 pixels across. If it looks tiny, lower the camera or use 1920 x 1080.

C. Teach the board with learn mode (15 min)

You teach exactly as an operator would on the line. Keep hands out of the picture every time you press CAPTURE.

1. Go to **http://127.0.0.1:8080/**. Type order **W900001** (or hold the printed W900001 label under the camera) and your initials, then **Next**.
2. Model step: type **PY-BENCH-01** and press Enter, or hold the model QR under the camera. It says the model is new. Click **LEARN MODE**. Description: **Bench test board**. Product line: **Bench**.
3. **Empty board:** take all 5 parts off (leave the model QR on) and click **CAPTURE**.
4. **Part 1:** put the **black** bracket (the usual one) in its outline, hands clear, **CAPTURE**. A bronze box appears around it. If it doesn't cover the whole part area, **drag a new box** on the picture (for the bag, box the whole pocket). Type the name and part number from the table below. Check the **Material** it suggests (Black) and set **Also comes in** to **White**, then **SAVE PART**. If nothing was found automatically, just drag the box yourself.
5. Repeat for parts 2-5, in order. For the bolts, put **all 6 down at once**, set **Layout** to **Several loose parts (counted)** and quantity **6**. (Fasteners in a fixed matrix, e.g. 24 nylocs, use **Grid** with rows x columns instead: the screen then names each empty cell.)
6. Click **ALL PARTS LOADED**.
7. **Reference 2 of 3:** lift every part off and put it back in its outline as an operator would, then **CAPTURE**. Repeat for **Reference 3 of 3**. This teaches normal placement variation.
8. Click **SAVE BOARD**, then **OK**. The station goes straight to verifying the board.

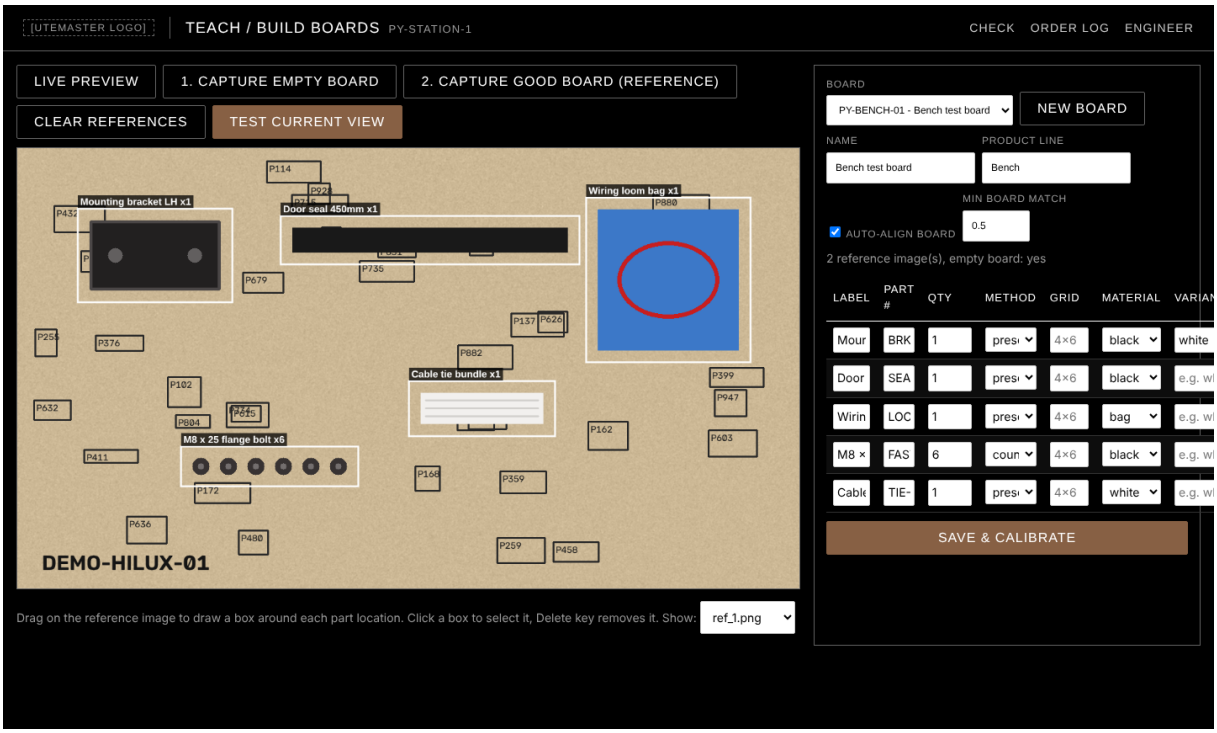
#	Name to type	Part number	Layout / qty	Material	Done
1	Mounting bracket LH	BRK-0412	Single	Black. Also comes in: White	[ ]
2	Door seal 450mm	SEAL-EPDM-450	Single	Black	[ ]
3	Wiring loom bag	LOOM-RR-02	Single	Bag (the camera may suggest Coloured; choose Bag)	[ ]
4	M8 x 25 flange bolt	FAST-M8-25	Loose, 6	Black (Steel if zinc)	[ ]
5	Cable tie bundle	TIE-200-WHT	Single	White	[ ]



Learn mode naming a part: live camera (left), what's taught so far (middle), the new part in the bronze box, and the suggested material pre-selected (right).

Check the teaching (engineer screen)

1. Open **http://127.0.0.1:8080/teach** and pick **PY-BENCH-01**.
2. Each part should have a box tightly round it, with the right quantity, method and material. The bracket's **Variants** column shows **white**.
3. Click **Save & calibrate**. The note under the button should say each part "looks" the colour you expect (e.g. "Door seal 450mm: looks black, material black"). A line starting with **!** means a material doesn't match what the camera sees: fix the material or the lighting.
4. With the full board in place click **Test current view**: every line should start **OK**.



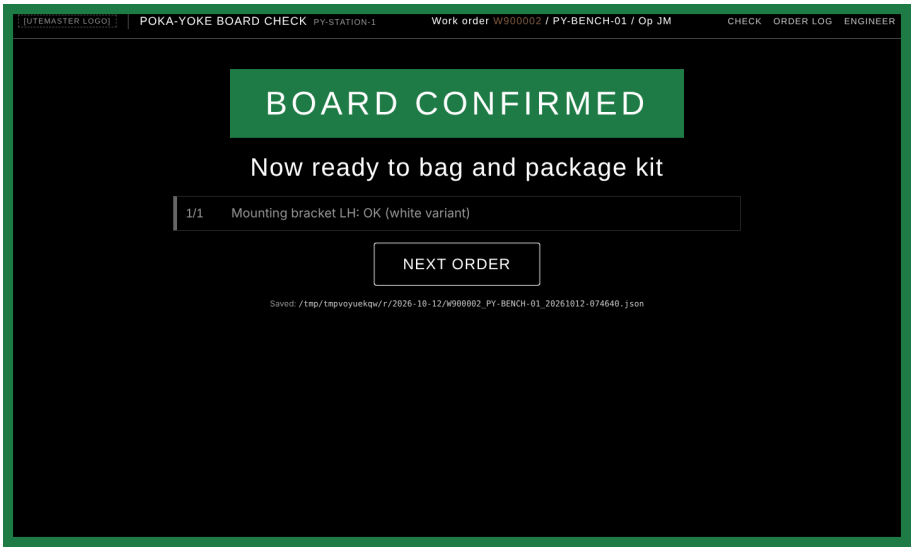
Engineer screen: part boxes on the taught board, with a Material column per part.

D. Bench test script (45 min)

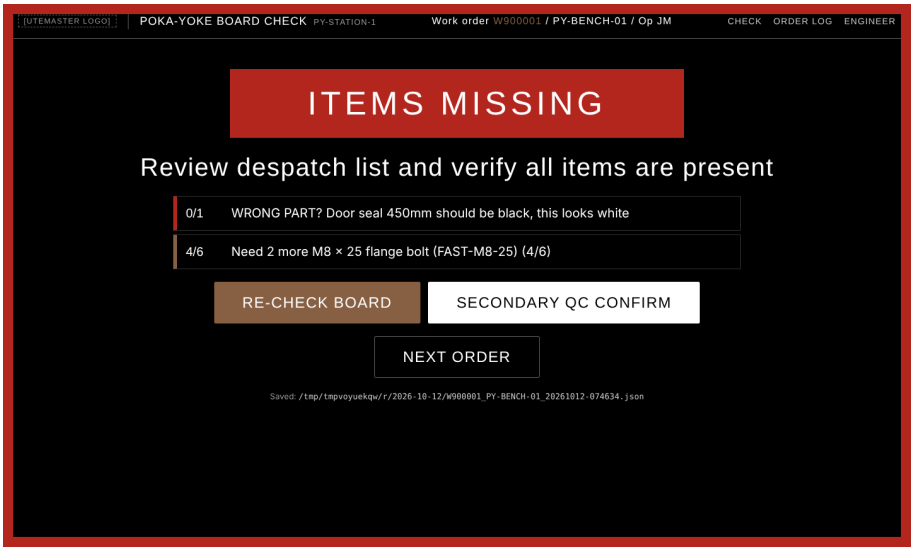
For each test: start a check (order **W9000xx**, model **PY-BENCH-01**, **VERIFY BOARD**), set up the board as described, wait 3 seconds, compare the screen with **Expect**, press **CONFIRM & SAVE** (or let it confirm itself), then **RE-CHECK BOARD** or **NEXT ORDER**. Put all parts back after each test.

Test	Do	Expect on screen	Pass
T1	Full board, all 5 parts	BOARD CONFIRMED, green border, "Now ready to bag and package kit". Confirms by itself after about 2 s	[ ]
T2	Remove the bracket	ITEMS MISSING: MISSING 1 x Mounting bracket LH	[ ]
T3	Remove the seal	MISSING 1 x Door seal 450mm	[ ]
T4	Remove the bag	MISSING 1 x Wiring loom bag	[ ]
T5	Remove the ties	MISSING 1 x Cable tie bundle	[ ]
T6	Only 4 bolts	Need 2 more M8 x 25 flange bolt (4/6), bronze line	[ ]
T7	7 bolts (add the spare)	Remove 1 x M8 x 25 flange bolt (7/6)	[ ]
T8	White strip in place of the black seal	WRONG PART? Door seal 450mm should be black, this looks white	[ ]
T9	6 silver screws in place of the black bolts	WRONG PART? M8 x 25 flange bolt should be black, this looks grey (if you taught zinc bolts as Steel, swap in black ones instead)	[ ]
T10	White bracket in place of the black one	BOARD CONFIRMED, with the line "Mounting bracket LH: OK (white variant)"	[ ]
T11	Bracket removed, rest your hand in its outline	Never BOARD CONFIRMED. Bracket shows MISSING or WRONG PART	[ ]
T12	4 bolts plus a fingertip where bolt 5 goes	Still Need 2 more (4/6). The fingertip isn't counted	[ ]
T13	Room lights off, desk lamp only	Repeat T1 and T6: same results	[ ]
T14	Desk lamp moved low to one side (strong gradient / shadows)	Repeat T1 and T2: same results	[ ]
T15	Phone torch shone at the board from 30 cm, bracket removed	Never BOARD CONFIRMED. Bracket MISSING, or "Can't see ... glare" / CHECK LIGHTING	[ ]
T16	Cover the lens with your hand, then press CONFIRM & SAVE	Bronze CHECK LIGHTING or CHECK CAMERA. Saved as NO_RESULT, not PASS or FAIL	[ ]
T17	Slide the board 3 cm and rotate it about 5 degrees	Repeat T1 and T6: same results	[ ]
T18	Put a different board or a plain sheet of card under the camera	Wrong board / WRONG BOARD or board not in view	[ ]
T19	Seal removed, CONFIRM, then SECONDARY QC CONFIRM: first your own initials, then AB	Own initials refused: "must be a different person". AB: white border, QC CONFIRMED, "Physically confirmed by secondary QC (AB)"	[ ]
T20	Next order. Hold the W900002 label under the camera, then the model QR	Beep, moves to the model step on its own, then loads PY-BENCH-01 with VERIFY BOARD ready	[ ]
T21	Close the black window, run run_station.bat again, open /log	PY-BENCH-01 still taught. Every check above is listed with order, result, photo and QC details	[ ]

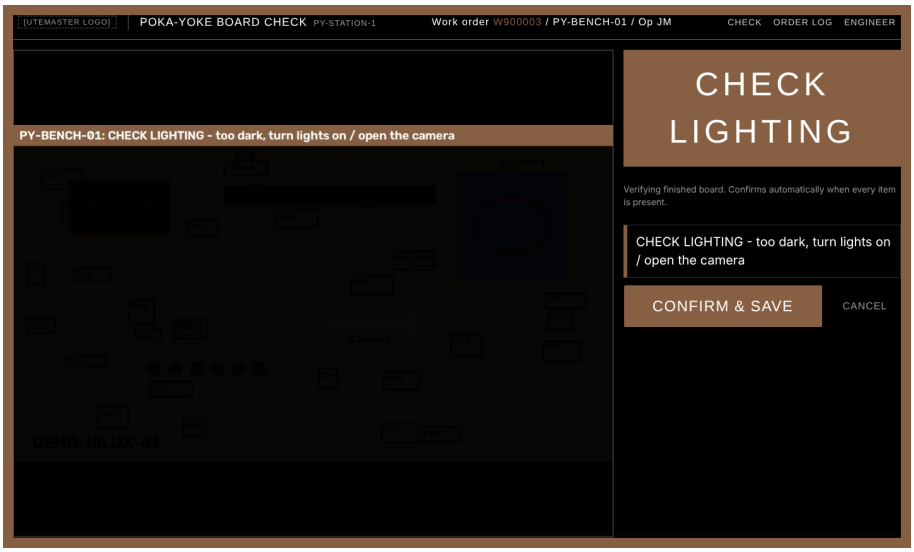
What the screens look like



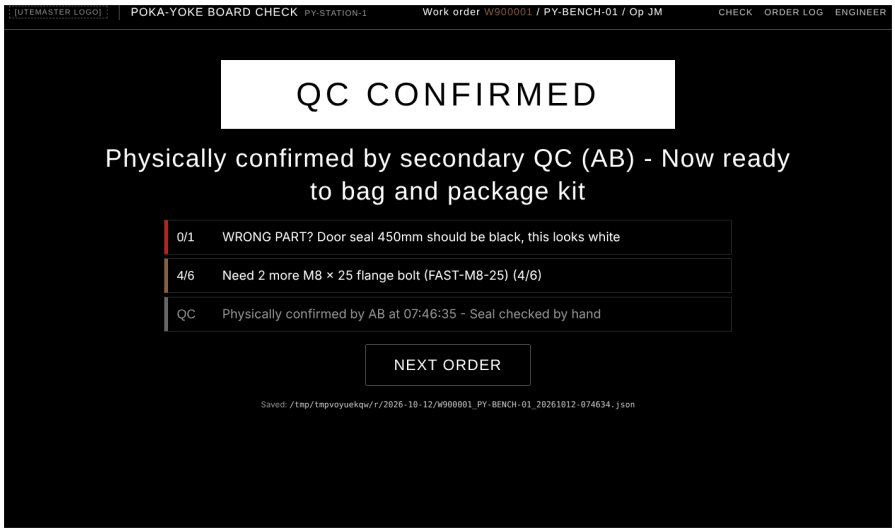
T1 / T10: board confirmed. Green border and the bag-and-package prompt.



T6 + T8 together: a wrong-coloured seal and 2 bolts short. Red border, each problem listed, and the option to re-check or call secondary QC.



T16: covered lens or no light. Bronze CHECK LIGHTING: the station won't pass or fail an image it can't see.

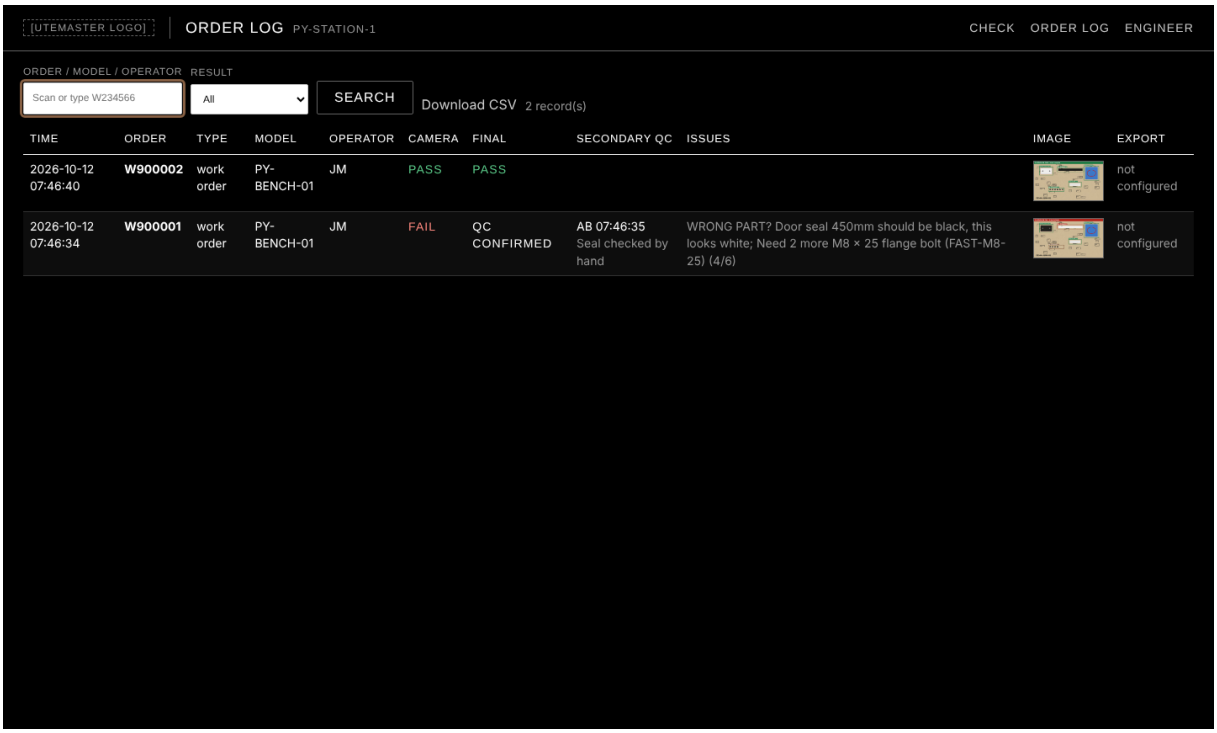


T19: secondary QC confirmed by a different person. White border; the camera result is still kept as FAIL in the record.

E. Results and sign-off (10 min)

Order log

1. Open **http://127.0.0.1:8080/log** (or **ORDER LOG** at the top of the screen).
2. Scan or type **W900001** in the search box: only that order's checks are shown.
3. Click a thumbnail: the marked-up photo opens.
4. Click **Download CSV**: it opens in Excel with one row per check (and a second row for the QC confirmation).
5. Result files are also in the station folder under **results\YYYY-MM-DD** (JSON + JPG per check).



Order log: every check with its order, camera result, final result, secondary QC, issues and photo.

Sign-off

Stage	Result (pass / fail / notes)	Initials	Date
A. Install + smoke test			
B. Camera + board set up			
C. Board taught, Test current view all OK			
D. T1-T10 parts, counting, wrong part, white variant			
D. T11-T18 hands, light, camera, position, wrong board			
D. T19-T21 QC, camera barcodes, log, restart			

Record any failed test with: the test number, what the screen said, and a photo of the board. The marked-up photo and JSON in results\ are the evidence.

Optional: launch from Ops Hub on this laptop

Only if Ops Hub (opshub.utemaster.com) is open on this laptop and the Poka-yoke step is published. Full checklist: **OPS_HUB_VERIFICATION.md** in the station folder, section 4.

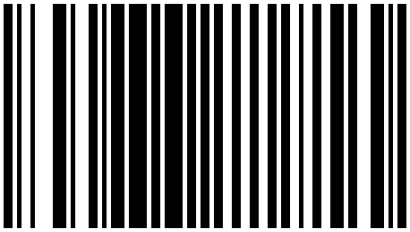
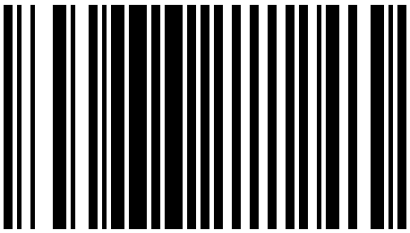
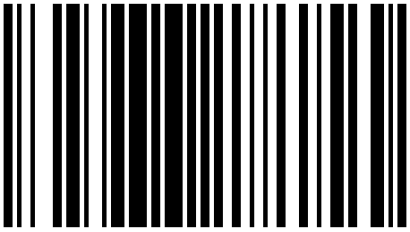
1. Close the station window. Double-click **install_windows.bat** once: it registers pokayoke:// links and starts the station at Windows login (hidden, no black window). From now on don't also double-click run_station.bat: the station is already running.
2. In Ops Hub, MES, Checklists: on a test checklist add a step of type **Poka-yoke Board Check** with Model override **PY-BENCH-01**.
3. Production Tools, Integrations, Poka-Yoke station: click **Test station on this PC**. If the browser asks to let Ops Hub access devices on the local network, click **Allow**.
4. Open the checklist on order **TEST123** and click **Launch Poka-Yoke check**. The station opens straight on the verify screen for TEST123.
5. Full board: BOARD CONFIRMED, then **Return to Ops Hub**. The checklist step shows PASS with the photo.
6. Repeat with a bolt missing: the step shows FAIL with "Need 1 more...".

Troubleshooting

Symptom	Fix
"py" is not recognised	Re-run the Python installer, tick Add python.exe to PATH, open a new command window
Install stops with a network / SSL error	Company proxy or firewall. Try off VPN or on a hotspot, or ask IT to allow pypi.org
Camera not found / black picture	Close Teams, Zoom, Camera app. Try camera_index 0, 1 or 2 in settings.json, restart
Picture is the laptop's own camera	camera_index is 0. Set it to 1 and restart
Parts not found or boxes in odd places when teaching	Drag the box yourself on the picture. Keep hands out of view; tape the board corners
Right board shows WRONG BOARD	Camera or board moved since teaching. Put them back, or re-teach. More marker lines on the board help
A material line starts with ! after Save & calibrate	That part doesn't look like its material under this light. Check the material or set it to auto
White ties flicker between OK and MISSING	Light board behind white part. Use a darker board or pocket, or move the lamp off the ties
Station is slow	Close other apps; leave fps at 2; 1280 x 720
Port 8080 already in use	Another copy is running (e.g. started by install_windows.bat at login). Use that one, or set "port": 8090 in settings.json and use http://127.0.0.1:8090
First install was interrupted, now errors on start	Delete the .venv folder in the station folder and run run_station.bat again

Printable test labels

Print this page at **100% / actual size**. Cut along the grey lines. Order labels: hold them up toward the camera (about halfway between board and camera) on the order screen; a hand scanner reads them too. The model QR goes flat on the board. Every label on this page has been checked to decode.

 **Order W900001 (Code 128)**	 **Order W900001 (QR)**
 **Order W900002 (Code 128) - test T20**	 **Order W900002 (QR)**
 **Sales order S900003 (Code 128)**	 **Sales order S900003 (QR)**
 **MODEL PY-BENCH-01 - stick on the board (50 mm QR)**	 **QC badge QC-AB: fills QC initials when held under the camera**